

Machine Learning Project Report

Predictive Modeling and Deep Learning Analysis on Passenger Datasets

1. Introduction

The goal of this project was to implement, evaluate, and compare traditional machine learning pipelines alongside a deep learning architecture using structured tabular data. Using data exploration and processing techniques, the pipeline aims to discover underlying predictive interactions, manage feature gaps, and contrast classical statistical approaches with fully connected neural network architectures.

2. Dataset Description

The dataset contains multiple demographic, socio-economic, and structural variables split into distinct feature spaces. Crucial variables include historical features, continuous price distributions, structural categoricals, and class targets. Preprocessing procedures involved addressing null values across columns, applying target/label encodings for string categoricals, and setting up structured matrix dimensions for input arrays.

3. Models Used

Three traditional machine learning models were trained and tested using optimized multi-variable setups to establish comparative baselines:

- **Logistic Regression:** Provided a strong statistical baseline with a model accuracy of **71.51%**.
- **K-Nearest Neighbors (KNN):** Analyzed spatial distance metrics, yielding a final predictive accuracy of **67.60%**.
- **Random Forest:** Implemented as an ensemble decision-tree setup, reaching a predictive accuracy of **65.36%** within this iteration's hyperparameters.

4. Neural Network

To challenge traditional workflows, a fully connected artificial neural network (ANN) was constructed via Keras and TensorFlow. The deep learning architecture consists of:

- An **Input layer** dynamic to the standardized feature shape.
- A first hidden dense layer with **16 neurons** using **ReLU** activation, followed by a **Dropout layer (0.2)** to mitigate regularized overfitting.
- A second hidden dense layer with **8 neurons** using **ReLU** activation, followed by an additional **Dropout layer (0.2)**.

- A final **Dense output layer** with a single node utilizing a **Sigmoid** activation function to map binary probabilities.

The model was trained using binary cross-entropy loss functions and automated backpropagation gradients, reaching a final test accuracy of **67.60%**.

5. Results & Comparison

The consolidated validation metrics show that traditional regularized models can remain highly competitive and occasionally outperform deeper sequential architectures depending on dataset constraints and feature dimensionality.

Model Architecture	Validation Accuracy
Logistic Regression	71.51%
KNN (K-Nearest Neighbors)	67.60%
Neural Network (Dense ANN)	67.60%
Random Forest Ensemble	65.36%

Conclusion Matrix: Within this empirical testing cycle, **Logistic Regression** achieved the highest overall accuracy on the evaluation split at **71.51%**. The multi-layer Neural Network demonstrated equivalent performance to the KNN model, showcasing that simpler, mathematically interpretable linear structures can yield better generalization bounds when dealing with localized tabular data volumes.

6. What I Learned

This project demonstrated that advanced neural network configurations do not automatically guarantee superior performance over traditional mathematical models when applied to specific tabular contexts. Critical learnings include the supreme importance of structural data engineering, the value of robust model validation benchmarking, and the strategic trade-offs between hyperparameter search complexity and actual optimization gains. i used ai